```
from google.colab import drive
drive.mount('/content/drive')

# !unzip /content/drive/MyDrive/dataset_machineAI.zip -d /content/dataset
```

```
Mounted at /content/drive
```

```
!unzip "/content/drive/MyDrive/Colab Notebooks/dataset_machineAl.zip" -d "/content/"
```

```
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e57d98a6-6dfb-4818-945d-5244a359635b___YLCV_NREC 0226.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e582071e-d792-4e0d-8f8c-dd88723eb2bb___YLCV_GCREC 5152.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5a29ee7-08b3-443f-8efb-7b136d3395e5___YLCV_GCREC 5425.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5acbbc3-c89d-4851-a4c7-f0c21401d500___UF.GRC_YLCV_Lab 02688.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5ad1ec6-6c94-40ba-b5d0-28eb397376ea___YLCV_GCREC 5481.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5b25092-f409-496a-a6f7-5f5bfd70ba83___YLCV_NREC 2417.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5b39e29-e605-41fe-98ea-ba1e47f062b8___UF.GRC_YLCV_Lab 08475.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5b5d434-c09c-4839-828f-ed6c402b8b0d___YLCV_NREC 2501.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5b770ee-62f3-41c2-a552-933fa8912734___UF.GRC_YLCV_Lab 09594.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5bb19cb-cd08-4701-a439-4da0f0bb65c3___UF.GRC_YLCV_Lab 02741.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5c5ba2c-edd5-4e9c-af78-b2ac2c4f828d___YLCV_GCREC 5150.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5d3227a-7a15-473a-b398-b98823570d69___UF.GRC_YLCV_Lab 02459.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5d64d96-df3e-4677-9998-7a2822326b0e___UF.GRC_YLCV_Lab 03265.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5dd23a4-e086-42e0-88a1-2b1f5b3d6fb2___UF.GRC_YLCV_Lab 02104.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5ed7151-897c-41a7-9967-4bfbdbe859ce___YLCV_GCREC 2927.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5fd4586-d720-4e69-892c-9e5e68beee6b___UF.GRC_YLCV_Lab 01727.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e5ffffda-a26d-4bb1-8d73-5ca656e2ff22___UF.GRC_YLCV_Lab 01986.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e6138418-bf0e-41ab-9db3-dc3c065b04fc___YLCV_GCREC 2385.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e6205580-80a5-45db-85d0-a7da5ad62724___UF.GRC_YLCV_Lab 08531.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e65e592d-3b5f-47b7-a7bd-29f154d88d5b___UF.GRC_YLCV_Lab 02159.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e669346b-90a1-4a43-b436-48c8d03d37a4___UF.GRC_YLCV_Lab 02300.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e6704ebf-fa77-4b3c-a8ba-ff42e50a5b43___UF.GRC_YLCV_Lab 03084.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e676cb10-638b-4286-8409-c8993e505a90___UF.GRC_YLCV_Lab 09437.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e678d581-9909-40ab-b344-bca3f9e95e0d___UF.GRC_YLCV_Lab 03097.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e67eb2bb-960c-4532-872c-d595c6de5461___YLCV_GCREC 1988.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e69414b8-7822-4017-aa9c-0b1c65b29e80___YLCV_NREC 2105.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e69cada6-46f9-4da4-a406-5b0d43ba7936___YLCV_GCREC 2511.JPG
neAl/Tomato___Tomato_Yellow_Leaf_Curl_Virus/e6a38626-2ee5-4008-808b-97bb0f438bfb___UF.GRC_YLCV_Lab 09527.JPG
```

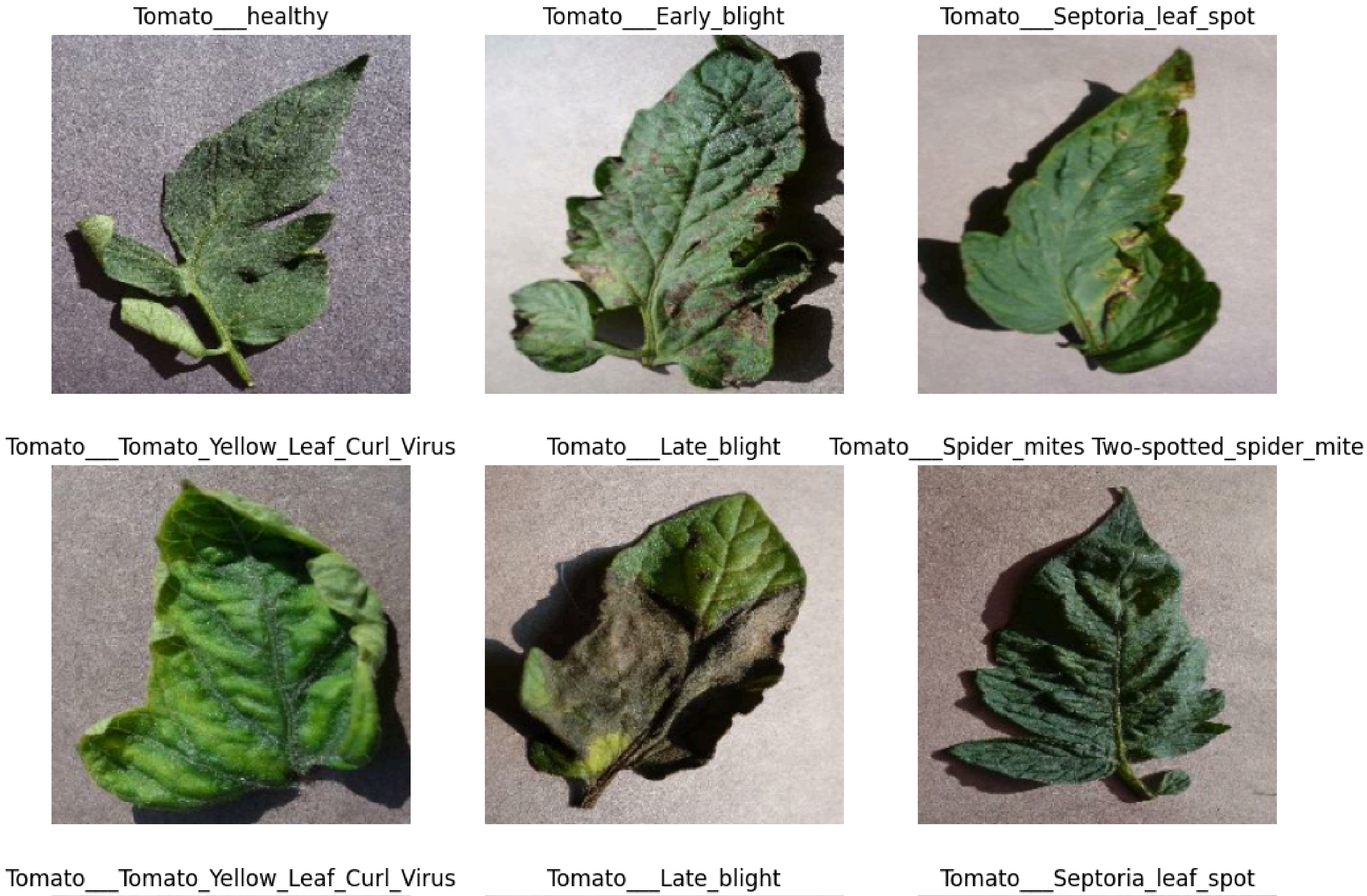
```
import tensorflow as tf

DATA_PATH = "/content/dataset_machineAl"

IMG_SIZE = (224, 224)
BATCH_SIZE = 32

train_ds = tf.keras.utils.image_dataset_from_directory(
    DATA_PATH,
    validation_split=0.2,
    subset="training",
```

```
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    label_mode='categorical'
)

val_ds = tf.keras.utils.image_dataset_from_directory(
    DATA_PATH,
    validation_split=0.2,
    subset="validation",
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    label_mode='categorical'
)

class_names = train_ds.class_names
print("\nDanh sách các lớp bệnh nhóm đang có:")
print(class_names)
```

```
Found 18160 files belonging to 10 classes.
Using 14528 files for training.
Found 18160 files belonging to 10 classes.
Using 3632 files for validation.

Danh sách các lớp bệnh nhóm đang có:
['Tomato___Bacterial_spot', 'Tomato___Early_blight', 'Tomato___Late_blight', 'Tomato___Leaf_Mold', 'Tomato___Se
```

```
import matplotlib.pyplot as plt
import numpy as np

plt.figure(figsize=(12, 12))
for images, labels in train_ds.take(1):
    for i in range(9):
        ax = plt.subplot(3, 3, i + 1)
        plt.imshow(images[i].numpy().astype("uint8"))

        # Lấy tên bệnh từ label
```

```
        label_idx = np.argmax(labels[i])
        plt.title(class_names[label_idx])
        plt.axis("off")
plt.show()
```




```
from tensorflow.keras import layers, models

base_model = tf.keras.applications.MobileNetV2(
```

```
    input_shape=(224, 224, 3),
    include_top=False,
    weights='imagenet'
)

base_model.trainable = False

model = models.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dense(128, activation='relu'),
    layers.Dropout(0.2),
    layers.Dense(len(class_names), activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

model.summary()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_we
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Model: "sequential"
```

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 128)	163,968
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

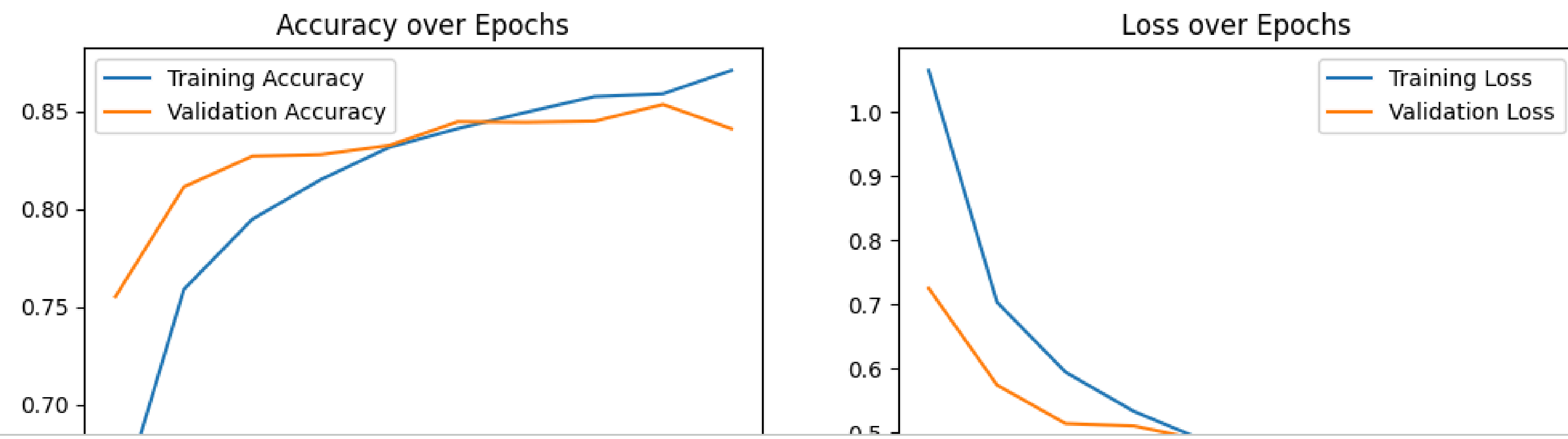
```
Total params: 2,423,242 (9.24 MB)
Trainable params: 165,258 (645.54 KB)
Non-trainable params: 2,257,984 (8.61 MB)
```

```
print("Đang bắt đầu huấn luyện... Nhóm đợi xíu nhé!")

EPOCHS = 10

history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS
)
```

```
bắt đầu huấn luyện... Nhóm đợi xíu nhé!
 1/10
54 ━━━━━━━━━━━━━━━━━━━━ 59s 85ms/step - accuracy: 0.6415 - loss: 1.0641 - val_accuracy: 0.7552 - val_loss: 0.724
 2/10
54 ━━━━━━━━━━━━━━━━━━━━ 50s 48ms/step - accuracy: 0.7589 - loss: 0.7031 - val_accuracy: 0.8114 - val_loss: 0.574
 3/10
54 ━━━━━━━━━━━━━━━━━━━━ 20s 44ms/step - accuracy: 0.7949 - loss: 0.5942 - val_accuracy: 0.8271 - val_loss: 0.514
 4/10
```

```
54 ━━━━━━━━━━━━━━━━━━━━ 21s 45ms/step - accuracy: 0.8151 - loss: 0.5330 - val_accuracy: 0.8279 - val_loss: 0.510
 5/10
54 ━━━━━━━━━━━━━━━━━━━━ 42s 47ms/step - accuracy: 0.8316 - loss: 0.4888 - val_accuracy: 0.8326 - val_loss: 0.488
 6/10
54 ━━━━━━━━━━━━━━━━━━━━ 22s 48ms/step - accuracy: 0.8412 - loss: 0.4547 - val_accuracy: 0.8447 - val_loss: 0.439
 7/10
54 ━━━━━━━━━━━━━━━━━━━━ 21s 46ms/step - accuracy: 0.8495 - loss: 0.4253 - val_accuracy: 0.8444 - val_loss: 0.444
 8/10
54 ━━━━━━━━━━━━━━━━━━━━ 21s 47ms/step - accuracy: 0.8576 - loss: 0.4119 - val_accuracy: 0.8450 - val_loss: 0.442
 9/10
54 ━━━━━━━━━━━━━━━━━━━━ 21s 45ms/step - accuracy: 0.8590 - loss: 0.3982 - val_accuracy: 0.8535 - val_loss: 0.422
 10/10
54 ━━━━━━━━━━━━━━━━━━━━ 21s 47ms/step - accuracy: 0.8709 - loss: 0.3651 - val_accuracy: 0.8411 - val_loss: 0.444
```

```
plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy over Epochs')
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss over Epochs')
plt.legend()

plt.show()
```




```
# Lưu vào bộ nhớ tạm của Colab
model.save('tomato_disease_model.h5')

# Hoặc lưu trực tiếp vào Drive của nhóm (nếu đã Mount Drive)
# model.save('/content/drive/MyDrive/tomato_disease_model.h5')

print("✅ Đã lưu mô hình thành công!")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`.
✅ Đã lưu mô hình thành công!
```

```
import numpy as np
from google.colab import files
from tensorflow.keras.preprocessing import image

def predict_tomato_disease(model, img_path, class_names):
    img = image.load_img(img_path, target_size=(224, 224))
    img_array = image.img_to_array(img)
    img_array = np.expand_dims(img_array, axis=0)
    img_array /= 255.0

    prediction = model.predict(img_array)
    index = np.argmax(prediction[0])
    confidence = np.max(prediction[0])
```

```
    plt.imshow(img)
    plt.title(f"Dự đoán: {class_names[index]}\nĐộ tin cậy: {confidence*100:.
    plt.axis('off')
    plt.show()

uploaded = files.upload()
for fn in uploaded.keys():
    predict_tomato_disease(model, fn, class_names)
```

Choose Files Screenshot… 201751.png

Screenshot 2026-05-12 201751.png(image/png) - 129709 bytes, last modified: 5/12/2026 - 100% done

```
Saving Screenshot 2026-05-12 201751.png to Screenshot 2026-05-12 201751.png
1/1 ━━━━━━━━━━━━━━━━━━━━ 13s 13s/step
```

Dự đoán: Tomato___Late_blight
Độ tin cậy: 100.00%

```
rt numpy as np
 sklearn.metrics import classification_report, confusion_matrix
rt seaborn as sns
rt matplotlib.pyplot as plt

ue = []
ed = []

t("Scaning")

images, labels in val_ds:
preds = model.predict(images, verbose=0)

y_pred.extend(np.argmax(preds, axis=1))
y_true.extend(np.argmax(labels.numpy(), axis=1))

ue = np.array(y_true)
ed = np.array(y_pred)

 báo cáo đầy đủ
t("\n--- BÁO CÁO CHI TIẾT 10 LỚP BỆNH ---")
t(classification_report(y_true, y_pred, target_names=class_names))

 Confusion Matrix
 confusion_matrix(y_true, y_pred)
figure(figsize=(12,10))
heatmap(cm, annot=True, fmt='d', xticklabels=class_names, yticklabels=class_
xlabel('Dự đoán của AI')
ylabel('Thực tế (Ground Truth)')
title('Ma trận nhầm lẫn - Đồ án Tomato Disease')
show()
```

```
Scaning

--- BÁO CÁO CHI TIẾT 10 LỚP BỆNH ---
                                          precision    recall  f1-score   support

             Tomato___Bacterial_spot           0.97      0.80      0.88       417
               Tomato___Early_blight           0.76      0.49      0.59       197
                Tomato___Late_blight           0.82      0.80      0.81       400
                  Tomato___Leaf_Mold           0.74      0.86      0.79       190
         Tomato___Septoria_leaf_spot           0.67      0.77      0.72       351
Tomato___Spider_mites Two-spotted_spider_mite       0.83      0.83      0.83       333
                Tomato___Target_Spot           0.66      0.81      0.73       288
      Tomato___Tomato_Yellow_Leaf_Curl_Virus       0.95      0.94      0.95      1070
         Tomato___Tomato_mosaic_virus           0.68      0.76      0.72        68
                    Tomato___healthy           0.94      0.95      0.95       318

                            accuracy                               0.84      3632
                           macro avg           0.80      0.80      0.80      3632
                        weighted avg           0.85      0.84      0.84      3632
```